

PEDIATRICIAN SCHEDULING AT BRITISH COLUMBIA WOMEN'S HOSPITAL (B)

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You log off the call with Meg and your classmates, ideas swirling in your head. There was a lot of information to decipher and now you need to figure out how to translate all of that into an optimization model. Just then, another email from Meg lands in your inbox.

MEG'S FOLLOW-UP EMAIL

Thanks so much for inviting me to your class – you all had such great questions and I'm excited to see what you come up with! I figured it would help to put down in writing some of the stuff we covered, so here goes:

There are 13 pediatricians ("peds") in the pool, including me. Peds are scheduled in three-week cycles, with each schedule typically released at the start of the previous cycle (i.e., 3 weeks in advance). Peds on call are assigned to either a day shift (7:30AM – 5:30PM) or a night shift (5:30PM – 7:30AM). That works out to 42 shifts per 3-week cycle.

Every shift must have exactly 1 pediatrician working it. There's a certain number of shifts that each pediatrician is supposed to work each cycle, and they also send me their requests for shifts they don't want to be scheduled for (I put it in an Excel file, attached). Peds can't work back-to-back shifts (i.e., no one works 24 hours in a row), and I can't schedule them to work consecutive night shifts (i.e., someone scheduled to work Tuesday night can't be scheduled for the Monday night or Wednesday night shift). And peds can't be scheduled to work more than two shifts in a given weekend (we consider weekends to be Friday night, Saturday day & night, and Sunday day & night).

We just started something called Pediatrician of the Week (POW). Essentially one person serves in a lead role for the week (Monday through Sunday). Every week, one pediatrician has to be designated as the POW, and the POW must cover the Tuesday day shift, Thursday day shift, and Saturday night shift. They can cover more shifts than that, but those three shifts are the minimum requirement for a POW. No one can be a POW more than once in a three-week cycle, and there's a certain number of times they need to be assigned as the POW for the entire year – if they've already completed their yearly requirement, then they shouldn't be assigned as POW for any week in the current planning cycle. I noted that in the Excel file too.

All of those things are non-negotiable since they're hospital policy or based on things outside my control. For example, a pediatrician's seniority, training, research versus clinic time, etc., will dictate how many shifts they need to work in the next cycle. That's nice in a way, since I don't have to consider the financial implications of the schedule, only whether it's doable.

But wait, there's more!

Peds prefer not to work consecutive weekends, and prefer not to work night shifts two days apart (e.g., if someone works Monday night, then ideally they wouldn't be scheduled for the Saturday night two days prior or the Wednesday night coming up). Along the same line, they prefer not to work more than two evening shifts in a given work week. My colleagues expect there to be equity in day/night shifts, so ideally each of their workloads would be evenly split between day and night shifts during the cycle.

Phew, I think that's all. Look forward to seeing what you come up with!

Meg

MAKING SENSE OF THE PROBLEM

After reading the email a few times, you know you need to make a plan before diving into programming a solution. Formulate first, then solve!

But something stands out: Meg didn't say if she was trying to minimize or maximize anything... she just wants a feasible schedule that adheres to hospital policies and satisfies her colleagues' preferences to a certain degree. It wouldn't be good if pediatricians didn't get their preferred shifts, but it wouldn't be violating any workplace scheduling rules if someone did have to work consecutive weekends, for example. Then again, if too many preferences are overlooked, it would probably make Meg's life difficult, and then this optimization model wouldn't be that useful compared to her current manual process.

Should that be the objective then, to minimize number of weekends pediatricians are scheduled to work? Or to minimize the number of times pediatricians are scheduled to work two evening shifts in a particular week? Can both of those things be minimized at the same time...? And what about equally splitting the day and night shifts across all the pediatricians? Normally, optimization models have just a single objective...

GOAL PROGRAMMING

You do a bit of searching and stumble upon a technique that's useful when there are several objectives, or *goals* that need to be met, called *goal programming*. It sounds like it will fit perfectly here, so you read up on it a bit more before you start your formulation.